

ActInf GuestStream 059.1 ~ David Tuckett, "Conviction Narrative Theory"

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hello and welcome everybody it is October 18th 2023 and we're here in active inference guest stream 59.1 with David Tuckett on conviction narrative Theory so David thank you for joining very much looking forward to this presentation and discussion so to you so thank you very much just perhaps a bit of useful background is that uh uh I'm someone who started out as an economist in the days when economists were taught more than they are now that is to say also sociology even a bit of politics and then I became a medical sociologist while at the same time becoming a psychoanalyst which may I won't attempt to explain but it's obviously different and then um in more recent years I've tried to put those things together with going back to economics and trying to understand particularly financial markets but more generally uh decision making and how it works are sort of on the big scale so conviction narrative theory is the outcome of this kind of exercise uh I suppose it's it most immediately started from the fact that uh I began uh a series of interviews but perhaps I'll just come to about that in a minute so what is conviction narrative Theory the purpose of it is that it tries to characterize the social and informational context in which decision making occurs and secondly the cognitive and affective processes governing it so it's about how people actually take decisions not how they ought to this book here minding the markets was a book I uh published in 2011 and was the outcome of 52 interviews with fund managers those are people who work for some of the large firms like Goldman Sachs or others and were responsible for investing at least 500,000 uh dollars into the world stock market some of them invested as much as 20 20 billion so they ranged and when I was talking to them I became aware that the they were very intelligent people they often had up to 20 people supporting them who are also very intelligent and able they had lots and lots of computer power and programs and analytic schemes and so forth but when it came down to it what they had to do was to make a judgment and they had to be convinced about that judgment and not only that they had to hold to that judgment over time which is something that's not often discussed in decision making so if you if you decide to sell

or buy or even hold any kind of investment that's a decision you make at $t=0$ but usually it's not going to pay off for you or the whole idea is it won't pay off for three five 10 years and so you've got to be willing to stick with your decision for quite a long period of time or if you decide you've got it wrong to back out and all of this requires what I realized requires what I call conviction so that's the basis of where this Theory started from so real world decision making which is what we're trying to understand here uh is best understood by example for example the one I've just given how to manage an Investment Portfolio but also a question like what level of reserves does a bank need to be safe a decision like should you expand your company into a new product or new technology or how much funding should a university or a company allocate to cyber security threats what resilience standards should be adopted by government regulation precisely what should we prepared to do to prevent catastrophic climate change how should we anticipate and prevent future potential financial crisis or how do you level up uh regions of your country or the world which are currently unequal these are all pretty big big questions and I would argue all of them involve uncertainty so they're massively consequential choices that's the first point the data is always going to be incomplete the options are ambiguous and the future that is what you know has happened so far not only may not resemble the past but in our modern world World which is changing rapidly is actually unlikely to resemble the past or you could put it another way which bits of the past will it resemble and which which it won't for all these reasons the axioms of standard decision Theory are not satisfied in this situation now the choices that require to make choice in this situation require both commitment and monitoring so as to adapt smartly over time as events unfold so in the investment example I began with you have to be willing not just to make the decision but to stay with it it's a bit like a marriage or any other kind of long-term choice of course you can get out of it but the point of doing it is to try and stay with it and then it'll also involve monitoring that is trying to assess if in fact you've made the wrong decision so you adapt pretty quickly over time as in events unfold now the context for such choices can be labeled as radical uncertain radically uncertain because data may or may not be useful and could even mislead now we should be able to say just uncertain or even we should be able to say risky but the problem is these words uncertain uncertainty risky have all become given very specific meanings uh in different disciplines which makes it extremely confusing but the underlying Point here is that I'm talking about a situation where the future is yet to be made and so it must be imagined however you're going to do that okay now conviction narrative Theory which tries to address this situation is designed as an alternative cross-disciplinary

theory of decision making that starts from the proposition that human cognitive affective and social capacities are well adapted for uncertain context a great deal of the existing research at least in Psychology has tended to uh demonstrate that humans are not very good at decision-making do all kinds of experiments which show that they have biases they frame things incorrectly they don't use statistics properly uh they get emotional Etc but by the time you finish reading all that research you wonder well how do we get where we are you might be you might be a bit say yeah well look where we are but if you look at the advances that that has been made by humankind clearly we're rather good at making new decisions and at developing new things and there must be something wrong with a theory which fundamentally talks about bias and one of the ways of thinking about that is to think that perhaps we're adapted to the real world better than we are adapted to the kind of world we set up in Labs or the kind of gambling decisions that typically uh decision-making Theory tries to start with so in both labs and gambling decisions there is a right answer or at least there's an answer that consensually people would tend to say was the best the best choice to make so what conviction narrative Theory does and I'll come to it in a moment is it turns the core issue from how to make best decisions which is typically what search has been looking up to now to how to make any decisions at all given the need to act when in fact there's a possibility of facing loss or failure real loss or real failure real disasters so how do people do it when they can't so to speak be certain or feel there's a really good chance of things working and the answer is that they adopt a conviction narrative and I'll try to explain that but the basic idea is they adopt of course a course of action which they believe to be sorted uh supported by a narrative which convinces them that by adopting that course of action they will get to the outcome they want now on the whole people don't invent conviction narratives just out of thin air what they actually do is they adapt narratives that is ideas about what actually are effective for the particular thing they're trying to do from those already circulating in their networks could be in their company in their social group Etc now one of the key things that conviction narrative is theory is designed to direct you towards is that if you accept that the basic way people make decisions is by trying to adopt a conviction narrative because of the need to act then what becomes interesting is how do they do this and in order to begin thinking about that the theory postulates two particular states of people or of organizations that you come back to in which this conviction is generated one is called divided and the other integrated I'll try and explain them more detail but integrated basically means where people bring all the information available uh to bear and they're as sensible as they can about it they realize what they're doing

they know it's uncertain but they're still able to make major commitments so it's kind of the academic Nana if I should make decision making divided is where for one reason or another people only consider part of the story part of the outcomes Etc and what conviction narrative theory is designed to do is to kind of alter the way way we approach these problems and so identify different kinds of ways uh for doing research okay now a central a central idea that follows from all this is the idea of ambivalence and this is an idea that I do get from from psychoanalysis so it's also used otherwise and ambivalence is an intrinsic property of any human relationship if you consider a marriage an employment contract or just about anything which lasts for any length of time you will tend to have at any given time feelings to in favor of doing it and feelings against doing it it's pretty normal that is feelings generate approach or avoidance but of course there'll be an overall level at any one time of one or the other once you get avoidance you get out of it once you have approach and you basically go on doing it and this is based on work going back to Freud or the sociologist Neil smelson now decision making in radical uncertainty necessarily evokes causes people potentially to be ambivalent because while time is unfolding once you've made the decision the outcome of your actions continues to be uncertain an exanti you quite simply cannot know the answer and that inevitably creates anxiety and anxiety creates avoidance so there's always a tendency to want to stop what you're doing perhaps or if it's the other way around and you decided not to do something the argument that maybe you should have can can come back so consequently contemplating action to exploit a perceived opportunity must stimulate a greater feeling of approach than avoidance pretty simplistic Theory but in other words approach feelings generate lots of let's do it kind of feelings and avoid avoidance feelings trigger loss aversion inhibition and withdrawal or a defense against that I just being blind conviction narrative Theory CNT Focus attention really on how this ambivalent situation which I want to stress is both affective and cognitive is dealt with and as I mentioned before the divided State or the integrated state are two ways of thinking about it you could think that if you accept radical uncertainty then most decisions really are experiments and if you play with that idea it's quite interesting because making decisions in so to speak experimental mode is rather different than making decisions in in other modes because if you're doing it in experimental modes it mean that from the start you would perhaps build into the situation as you do with an experiment ways of measuring what's going on giving you feedback formalizing your hypothesis as to why you think things are going to work out and then uh uh getting some sort of feedback as to what's actually happening now obviously one trick here the trick that's but which most

disastrous potentially is you simply eviscerate doubt so that uh and one of the problems is that a lot of modeling decision a lot of modeling that is called for in public decision making is often used whatever the modelers might have intended to actually get rid of D oh my model tells me it's all right to do it sort of thing okay so to say a little bit more I've used the term narrative and and so it's worth trying to say what is actually meant by that in our Theory narratives are structured higher order mental representations which incorporate all of causal temporal analogical and veilance information about agents and events which serve to explain data imagine and evaluate possible Futures and motivate action so it's their quite a lot of things that look like narratives which aren't narratives unless they have the potential to do all those things that I've I've just read out it it may not always be in the narrative but it is there potentially for it to be a narrative so you know you would tell a story about what would be the effect of giving this talk for example now narrative fragments are subsets of the elements in a narrative which can be readily communicated so it doesn't seem to us after a lot of thinking about it and study that there's very much to be gained from uh trying to decide what when is it a big narrative when it's a small narrative when it's a narrative when it's a story to have to recognize that narratives tend to be made up of other narratives of smaller fragments on which the main n narrative depends the second important concept is shared narratives uh what I'm getting at here is that uh there are narratives that are shared within social networks which are so to speak do dominant so in a particular group this is the way we do things this is the way we understand things science is lots of shared narratives which govern scientific methodology for example third important term is explanatory models this is taken from social anthropology and explanat models are culturally valid valid systems of understanding often ex expressed as narratives so for example a theory of economic growth is explaining how economic growth takes place a theory of coid infection and so on finally The crucial a crucial concept is feelings and this is based on the idea that feelings and modern neurosciences demonstrated this pretty strongly that feelings are at the center of all brain activity that is most things you can observe when you look at brain you know brain scans and things like that you typically see Mo movements between different parts of the brain and often going through uh parts of the lower brain uh Etc the amydala and areas of that kind the function of feelings it seems is that they force their way way to Consciousness in the human brain and they prioritize conflict resolution in context and this is particularly important it's especially important for economists who for example have a a theory of utility because I think what we learn from uh what we know about feelings in the brain is that your utility function Alters according to context and

the brain is sort of set up to do this so we actually have a number of conflicting a which where one or other is turned on according to the particular context for example uh to look after people caring as opposed to enjoy ourselves and so on and so forth the overall effect is we feel we want to approach or avoid uh including when you're listening to me some of what I'm saying may be sort of switching you off completely where you may be struggling to listen or even not want to listen at all other bits may be making you feel yes yes and this is going along with whatever cognitive judgments you're making as well so to summarize I don't see although it's useful to talk about cognition and feelings maybe separately they always operate together uh that's a picture of the brain to show the point I made which I won't develop now but the the fundamental point is that for a long time in the 19 Century the brain was understood the wrong way around that is to say just because the cortex is on the top there and information comes through the core actually it's the the lower parts of the brain which are fundamental to conscious decision making and the the uh the the cortical function is kind of added on top now there are four functions of conviction narrative theor which explain why conviction narratives work as they do this slide doesn't seem to work so the first function of a of a narrative in conviction narrative is explanation that the narrative imposes structure on the current situation that yields a sense of understanding and emotional satisfaction so for instance typically when people do science they do all the stuff they do with the methodology and so forth and they present results but in doing so they tell a story about it explaining the situation and that we do most stories have an explanatory uh quality the second element uh in a conviction narrative is it allows a mental simulation so for example if you are thinking about shall I um shall I adopt this particular way of trying to do climate mitigation I can think of the actions I can imagine I can get all sorts of spreadsheets and and you know proper hard information and I can try and look at what I think will be the outcome of that particular activity in 20 years time and as I do it in some way or other I'll be forming at least implicitly pictures in my mind of what the earth looks like when I finish doing all this and then you can compare an alternative or many alternative narratives and so to speak you run the narrative forward to generate imagined Futures as associated with a particular choice the third which follows from what I said about feelings is that these explanations and mental s simulations are evaluated effectively so you react emotionally to the Futures you imagine to evaluate their desirability and that helps to manage commitment over time and the fourth and really important function of narrative in a conviction narrative is communication most important big decisions of the kind I'm interested in involve a number of people working together it could be

several labs for instance or it could be people working to a particular aim in regard to climate or or many other problems and so when you work together with other people you have to motivate them and you have to explain to them and you have to give them an idea of where you're going and what the point is and we use narratives for that and narratives that propagate across social networks and U become in that way maybe shared narratives so the other thing I'd like to stress is the conviction narrative is not any old narrative it's a narrative that sets out how you're going to change the current situation by an action to lead to a desired outcome okay it's very spe and to do that you draw on narratives in your in your your in your context so the narrative solve uh two particular uh or or there are two sets of problems in decision making if you think about it we call this the logic of decision so that decisions reflect both data picked up from the external world including the social environment and internally derived goals so to make a decision you've got to do both things you've got to U pick up data and then add it to your to your goals the first aspect to this you can call the mediation problem which in the diagram is the dashed lines it reflects the need for an internal representation what in our paper we call a currency of thought that can mediate between data from the external world and actions decided internally and one of the things about economics for those who are economists is e economics actually has as far as I can tell no theory of cognition models in economic you just write down um you know R for real rates it doesn't have a theory of how people come to know what the real rates are and of course as soon as you do have to fix such a theory you've got lots of room for disagreement and differences second is the combination problem which is the the lines and that reflects the need for a process a driver of action something to get you actually doing something that combines beliefs and goals to yield actions and In classical Theory classical decision Theory the currency of thought is probability and the driver of action is expected utility maximization so you know if you've got a probability of 70% of success and and that that's good enough then what you're supposed to do is to choose the highest thing and that is a very seductive and uh sensible way to make decisions provided you think you've got probabilities in behavioral economics the main emphasis is on how people get the wrong probabilities they make decisions based on biases and so on that they don't take the probabilities properly into account in radical un certainty that the argument is there aren't probabilities or at least there aren't probabilities for the kinds of questions that I've been trying to talk about so in CNT the currency of thought we propose is narratives and the driver of action is effective evaluation of narratives in a social context and this a rather complicated slide to say something about about that you see that you know we have data we have the

narratives we have this explanation and simulation there's affective evaluation leading either to approach or avoidance link with your goals and then you take action so that representation and processes in conviction narrative Theory narratives supplied in part by the social environment there a very important part of the theory that we're picking up narratives from around us all the time are used to explain data they can then be run forward in time to simulate imagined future outcomes of action which are then evaluated effectively considering the decision makers goals and these appraisals of narratives then govern our choice to approach or to avoid these imagined Futures and there's a you see from the figure there's lots of feedback loops okay and I'll just talk a little bit about divided and integrated States so uh as I mentioned this diagram is supposed to uh illustrate uh narrative prediction that is what your narrative is going to tell you to do as you get new information based on where you started your prior so in so far as the new information you get is creates congruent feelings to where you were before you can pay attention to that data in an integrated State you pay attention whether it's congruent or not congruent with your existing narrative in a group Feld or divided State how it works is you only pay attention to the things that you already uh agree with in other words confirmation bias and in congruent Fe information that produces in congruent feelings is ignored so this this is how you get bubbles in finance and Bubbles in lots of areas of activity in society because if you don't if you don't attend to the negative information then that's the only way you can go now bear in mind this can also work the other way if for example you're in a depressed or in a in a state where we're not doing anything you can continue to there you discount the positive information and so you stay in a repetitive bubble so an integrated State group of Affair State of Affairs it is is that envisaged in normative theories of science based on experimentation and or inference strategic actors are curious a broad range of information and perspectives both supporting and challenging plans and actions are considered and it's recognize that there are uncertainties and contradictions within the preferred conviction narrative supporting a strategy or policy and crucial to the integrated state is the idea you know that although you're making a decision it could be wrong so you're open to new information The Divided State I won't go through it again is is just the opposite of that okay and this is just a a last slide to just give a quick uh link to work that's been done particularly in Psychology where you can see that this whole theory is very dependent on what's going on in your local social environment because in your loal local social environment there are various narrative elements influencing your evaluation of narratives so you know the the stories that are around the his the shared history of the group various characteristics or Logics or frames or conventions beliefs and so

on which are all uh but these interact with the three things which have been found very uh significant in in lots of psychology what are called content effects that is ways in which different content tends to be picked up positively or not trustworthiness effects that there are all sorts of effects about listening you know you can dress somebody up to look like a professor and people are supposed to believe them more as opposed to a or something uh and presentational effects which has to do with the way things are presented and so on this can influence whether a narrative is convincing or not and overall you get the the usual thing that It All Leads to a narrative which either generates approach or avoidance and only if there's a positive balance do you go for Action okay so that's roughly the theory I hope it makes some sense and uh uh generates some uh questions all right awesome okay well while I am cropping and returning everything to normaly visually how did you come to this I mean what brought you through the the warps and the weaves to want to highlight this perspective well so first of all I think when I was I was enough to study e economics at King's College Cambridge it been the College of John Mano KES and when I was there it was still very much we were taught by kan's pupils and they saw economics as very much to do with the management of uncertainty economics today has defined uncertainty in such a way that it doesn't have to take it into account and so everything can have a probability and so on so I I've always felt uncomfortable with what's happened in economics because the real world seems to me uncertain as I said earlier um you know there's the whole point about human Innovation and and the way in which new companies are formed in an economy or new scientific theories come about Etc is is there new they may be recombinations of the past but they are they're new and and it it's not helpful to have a kind of equilibrium theory for for those sorts of problem however useful it might be for other things and so that was a general background um it was through psychoanalysis that I came to see the importance of feelings uh most people when they think about psychoanalysis probably haven't don't know much about it or had a rather half-baked explanation given but actually what Freud's major contribution is uh was that he understood that so to speak feelings and thoughts are intricately linked that if the thoughts you're having make you feel bad you'll do all kinds of things not to have them or even not to need to have have them in the first place so this link between thoughts and feelings has always been around it actually came in part from his Neuroscience but in more recent times it's now uh become I think largely accept uh view I gave this Theory the outlines of this Theory to I tried out on Carl friston who probably most of you know about 10 or 11 years and Carl is a great listener he sat there and he listened for about 15 minutes and he looked very thoughtful and then he

said well there's nothing you've said which is in opposition to current Neuroscience so that was nice that encouraged me to go on anyway so uh also as as I've mentioned when I began trying to understand uh how could uh the do crisis in in financial markets have come about which I was studying just as the global financial crisis was actually not happening but about to happen it was then that I realized and I think quite a lot of people have realized it afterwards that this kind of thing in which people make decision based on convictions they do the best they can with the information they've got but by its nature that what they do is uncertain and therefore fragile awesome yeah definitely wh while there wasn't any active inference generative models in the talk you brought up so many motifs that I think we can explore a little bit because they're the kinds of things that that as you pointed towards necessitate a cognitive account so first how how would you see cognitive modeling playing a role in microeconomics or macroeconomics like what kind of cognitive model what kind of explanation prediction or intervenability do you seek well so I I think I mean there's there's no doubt that that um you know as we as we go about our business we form cognitive models of the world around us and um that in my opinion that's a useful way way way to think uh and i' I've really called them explanatory models because I first encountered them in anthropology this was when I did a big study in in the late in the mid 80s of um a thousand patients going to their doctors and looking at what they'd understood and to cut a long story short although at the time it was widely thought that patients didn't understand most of what their doctors said and therefore it wasn't really worth their doctors spending much time talking to them uh in fact what we found was that most patients did understand what the doctors uh were saying but that where they didn't it was because the model of understanding what was wrong with them that the patient had so they had their own model was not the same as the model the doctor had and they hadn't established the the two explanatory models and then you know put them together and in fact very often it would have been quite easy for the doctor if he knew what the patient was thinking to point out that this that or the other assumption or process as as we would say you know wasn't how things worked and then that would have helped the patient to understand as it was the patient just went away not quite understanding what had gone on and often didn't follow the treatment so that's how I got into if you like cognitive models now when you try to formalize them uh uh you know I think it's extreme it's extremely interesting to to try and formalize them but I always have the reservation that um you know where it's always where we're getting the information from and how robust is it to changing circumstances so I mean that that's the that's I think the key problem cool so I was very um interested by that

provocative example of the person making financial decisions with with a 20 person team that's a lot of attention and energy a lot of skin in the game just for that kind of an organization to to deploy that kind of a decision support setting so how do we think about that what what works or doesn't when there's group decision making and then how does that integrated divided dialectic within an IND individual relate to the status of the integrated or divided understanding that might arise at a group level especially if it's a decision where like it's obvious from the get-go that no no one of those people know whether the stock is going to go up or down so then what does that 20th person or the 21st person or the 22nd person do on that team to actually empirically lead to better outcomes that's a really it's a really good question uh so and in order to to to think around it there there certain background facts that are important one is that these people doing that work are on the whole the most highly paid people in our society which is which adds a level of Fascination to your question because we know we know from academic research that no one uh no there's no published data anyway to suggest that anyone can consistently do better than the average okay and the reasons are obvious uh nonetheless there's always hope right and so one of the ways you could start to think about it is is this uh supposing it's it's your P supposing it's your pension and you've got a choice to put the money in something which will just get you the average which of course will mean that at times it'll go down and times will go up ex you'll just get the average versus somebody who says well you know I can do better than the average look at my last year's results and they last year's results will show better than the average because obviously by by statistics some people do better than the average so then it's quite difficult for people it seems to say no to the possibility of doing better and that might be one part of the story here but at the level of the the people with 20 teams um they they essentially are are paid to do this work we know that they do it better than you or I could do do it on average so there there are studies that show that individual investors don't do as well as uh professional investors but they are increasingly studies that active investors that is people doing sort of what I you know who look over the data and try and decide what to do that active investors in in fund management don't uh on average beat some form of more uh computer-based automated way of doing it which just looks at averages and so on of course all of these methods fall apart under certain situations uh computer-based methods did terribly badly uh at certain times and they have done recently um so and we also know that if you so that if you know what private Equity if you heard of private Equity or Venture Capital this is another level which ordinary people aren't allowed to invest in because it's supposed to be risky but where actually they often do do

better though uh because they've got less money to play with and they can use expert knowledge in particular areas again it's subject to all the problems I've talked about so why what what is all this about and why are there all these people engaged in and I think the answer is that there's a constant attempt to support narrative that you know what do you do with your money so I mean the University's got University Pension funds got to invest right your pension it's got to be done somehow and uh uh yeah I could go on a great deal because I think the conclusion from this is that it doesn't work very well and that's because what people do is they pursue what I call in the book fantastic objects that is they become convinced that you know the new way of doing hydrogen technology or the new way of doing this that or the other which is a way which is promising but which nobody knows enough about to be sure about that's going to be the thing to make you your millions and that's what you'll invest in or there's ABC person who's a fantastic investor he becomes a fantastic object okay so that what a fantastic object works with group think in a divided state to get rid of doubt and then create this process I described earlier which is a sort of self-fulfilling prophecy process which works until it doesn't cool another piece I found really interesting is the laboratory decisionmaking setting whether it's reductionist or however you want to characterize it it focuses on certain kinds of decisions first off ones that align well with the axioms of standard decision Theory but just pragmatically they tend to be these oneshot gambling decisions or Game Theory decisions that are really amenable to the kinds of Matrix calculations and the nash game theory and all of this yeah and yet that is not how it feels to be making decisions day to day and then you brought in the approach aoid so when the gambling is in front of us and we only have the decision-making Alternatives it kind of removes this meta on the decision like whether one even approaches that slot machine or not which is a much more nebulous decision because everybody starts in a different position and then also this consistency which is really where conviction comes into play with an with the marriage or with the investing which is even a decision at one time point is really more like an agentic event in a broader tapestry or narrative whereas in the gambling setting it can lead one to make models that are sort of like decision next next next next and it's like blank slay every single time that a new decision is made and then that has limited ability to transfer in to all kinds of other situations that you describe as having that radical uncertainty like to the root rather than a superficial uncertainty like whether in this one narrow decision something should happen One Way Or Not yeah exactly I I think that's that's exactly right and of course the I mean it's not really to be negative about all the stuff that's been done because it's shown very interesting things but I think the the main the

main issue is have you identified your context correctly because we kind of know I mean in the to go to sort of basian stuff we as I understand it although I don't fully understand how it's really done uh there there's an argument that that we're good Samplers that we sample well locally people do on the whole but of course the problem comes when you get out of your context and so the the I think the key issue is sort of the transferability of information or Knowledge from one context to the other some friends of mine Lenny Smith and um Erica Thompson have written a paper called uh escape from modeland which is all about how you know they're modelers he he they do forecasting weather forecasting and climate models and all that but the question is there's lot you know how do you get out of the model into the real world and what challenges do you then uh face I mean and that's much more of course economic models are much weaker than those kind of models they're talking about so but it's a context thing and how can you how can you assess whether you've got the right things in your model you've got the right data Etc and these these are the things where you tend to I mean the financial crisis was partly caused by people having the wrong models but they'd seem right at the time yeah well the AC of sampling a good sampler is exactly what active inference is about are is secades are in the business of doing good sampling good epistemic value and when the context shifts like if you have a book and you flip it upside down your icting is not going to be making as good sampling and that's why it's hard to read let me exactly ask some questions from the chat all right Bert asks there are many similarities with scenario planning in transport policy how could policy planning move from cost benefit analysis to the use of narratives well that's that's a very that's a very good question there's a there's a paper on the second paper down that deals with Futures and foresight and conviction ality Theory but so the the problem with cost cost benefit analysis is kind of a logically sensible idea in which you know you try to work out the costs of um of of a project and you try to work out the benefits and then you work out whether it's worth doing the problem is with all the assumptions as obviously the questioner realizes and in a formal cost benefit analysis you have to make up lots and lots of numbers in order to make it formally work so I think the question is whether you're using the cost benefit analysis to have a discussion about making a decision and to open up ways of thinking and question assumptions and all that kind of thing or whether you're using it to come up with an answer which then you aren't responsible for and you can say you know the model told me this I mean putting it bluntly but obviously the major problem with with cost benefit analysis in transport is you have to you have to make all kinds of assumptions about the benefit of of you know going half an hour quicker or of having fewer

people in your in your uh Railway carriage or over what period of time you're going to do it we we know that most major projects have always overrun costs have always been heavily criticized at at the time uh the Sydney Opera House is quite a good example in a completely uh iconic building but was a total disaster when when it people first played music in it Etc and and obviously we have things closer to home in the UK at the moment which have this problem so I think where conviction narrative Theory can can help you with that process is by first of all helping you to look at try to dissect your narrative where are you getting your conviction from and which bits of the information you're getting which narratives are you know giving up are being considered compelling which narratives are not being discussed what points of view for example you know the workingclass people through whom the freeway goes and not being and not being taken into account um uh Etc um so I think it can assist with thinking around decisions and with using formal methods which I'm I'm absolutely not against using formal methodologies but using them properly and checking out they work in the context cool all right another question from the live chat so shagor writes even if on average generating Alpha returns are not possible is there work that identifies selected investors such as Warren Buffett and others that have consistently identified asset mispricing wouldn't identifying investors that are able to consistently spot mispricing support conviction narrative Theory I.E sophisticated or successful investors might understand how the narrative could change well so I you know I haven't done the research on uh uh on uh whether or not there are consistently success investors but as I understand the literature and it's pretty consistent um the answer is no there aren't any consistently successful investors uh and those who are successful eventually aren't I mean I happen to have as a quite a close friend of mine George Soros who is one of the people who did very well but he also failed if you talk to him he he he he failed and uh uh of course once you get like Warren Buffett it's complicated there because if Warren Buffett does something you know it other people do it and so but it's of course possible as all the statisticians on the um on the on the listening to this would tell there's no problem having a few outliers statistically it doesn't tell you that it's anything to do with those individual people I mean it might uh so conviction narrative theory is is is would be would explain how people manage to persuade themselves and each other that their active form of investment works and it would also help to be used to identify the basis of their claim um but on the whole I think um uh you can understand a great deal of what goes on in the F in the financial markets by deploying the concepts of fantastic object that I mentioned um integrated State and divided State because in an integrated State it's very difficult to do

investment very difficult indeed for the reason you said in your questions to me you know you said well obviously having 20 people in the room isn't necessarily help but you can convince yourself it helps and and I can tell you there are some incredibly imaginative people out there who work incredibly hard so one person I interviewed he had not only got his team of 20 but he'd assigned them different roles to to be for and against things to do this calculation other to play Jokers when uh they want wanted to make a particularly strong point and he paid them according to the way they behaved and so on they worked really hard to try and get it right but it didn't it didn't end up he was convinced but it it didn't end up over the long term being any better than any other way interesting so even from our ACM post we've seen some inroads into cognitive economics where people are drawing from the recent developments over the last decades in cognitive science the pragmatic turn extended embedded enacted and cultured affective all of these different features that are increasingly being understood in the context of human cognition as well as nonhuman cognition biological or synthetic or otherwise so where where does economics go as it also takes this cognitive turn how have you seen economics play a role from what you've worked on over the decades and then like how would it have been different if there would have been this cognitive turn earlier or how will it be different going forward as there's more of a engagement with economics and cognitive science so I think so I I think first of all microeconomics uh microeconomics in in its sort of best light uh has small models that tell um illuminating stories about the relationships between certain I mean simplest one is supply and demand um and it it tells useful stor is however the question is is that model working in any particular situation you're in so even supply and demand you know what if the I mean the model is still true but it may not explain the situation you're trying to deal with and it occurs to me while he was speaking actually one very interesting example so there is a there is a high level academic group that was uh set up to look radical uncertainty which has um has on it many people much more eminent than me on both sides of the Atlantic and on one occasion uh a very well-known American Behavior EC colist presented the work on gambling that you alluded to and gave the group sort of you know uh the same test that he'd given to his Harvard economics class earlier that week right he was completely astonished when this group of people didn't give the answers that he was expecting and he got out of his very intelligent Harvard graduate class and the reason was that people said things like well if that bet was available I know the bank wouldn't pay out and in other words they thought round the problem for things institutional constraints so that they didn't just think of the problem very narrowly framed whereas obviously the The Graduate students were being trained to

think narrowly which can be which can be useful right um so I I think where the the fundamental issue with economics since I studied it back in the late 60s is that it's become entirely nearly entirely modelbased there are of course exceptions but the the sort of high Prestige of the discipline is in producing a model and if you read the papers they're often like this let's approach it with a simple model we'll have one person you know and and they're always simple models there are always lots and lots of assumptions they um they they work very nicely but it's totally unclear how you apply them to the kind of world that I'm talking about meanwhile what economics doesn't do very much of is actually go out into the real world and actually look at what happens what people do so the study I did of fund managers you would have thought someone you know I mean I'm an amateur right you would have thought someone in financial economics would have done this a long time before uh but they haven't they they they write lots of paper they do lots of analysis of very large data sets and things like that but they don't actually look and see well how are people how are people doing it so I think one of my hopes would be that it would it's it's not that you don't want models but you need to understand a lot more about the human context in which they're being used and in particular I think uh you know typical discussions of Behavioral economics for example is you know you compare rational and irrational Behavior where rational means entirely modelbased right which wouldn't necessarily be rational in ordinary terms and irrational means anything else now if you can't apply probabilities the whole model doesn't doesn't really start I think that this restricts thinking about well how do how does emotion for example enter into decision making and and there's there are a lot of studies in the management literature of what happened to Nokia you remember Nokia were producing mobile phones and they were producing most of the nokia phones on the planet and then suddenly there weren't and this happened pretty quickly and that some of the arguments about it concern complex emotional issues in the company where the people who had done very well for Nokia over many years were essentially Engineers who built really good kit but they didn't understand and hadn't they were neither listening to the market so they weren't like the people trying to sell the things and nor were they software Engineers so they hadn't yet understood that actually the hardware doesn't matter much it's the software because you know if you compare many of those old NOK phones to an Apple I mean the nokas were engineering Superior things in many ways but the apple of course one with all the things it could do so um I I think you can you know you need to there need to be many more studies of this kind in my opinion yeah what you said there about there being Legacy prestige in presenting model-based local decontextualized

variability reduced analyses reminds me of ant foraging models where it seems like every single Paper can present a simple model of a single context making many explicit and implicit assumptions and then the loop is not always closed by going back out there so I guess my question is how do narrative models help us or what else do we need to supplement or complement potentially very sophisticated cognitive narrative modeling because if you point to that phenomena in the setting of matrix-based Game Theory modeling and people think that just by constructing that model that it's an open and shut case well now here's a more shiny and feature laad in type of modeling that seems to cut closer to the real Crux of the issue with sense making on the inbound and action selection on the outbound so how do we work so that cognitive modeling or narrative modeling doesn't also fall into that trap where people just present a simplified cognitive model and then continue the [Music] same kind of um epistemic Gestalt in how it's deployed so what else is needed beyond the model or what sits beside the model well so I don't know whether this is really a very good answer but but uh I've never been more than a part-time academic which is no doubt bad thing in some way but I I think the really fascinating so so you know models can can at least claim to give you a general truth and the problem about General truth is it may not apply to any specific situation right but it people many people find it elegant I actually find real interest in problem solving and particularly when you've got I mean I think interdisciplinary work is fascinating it's not fascinating when you all sit around and each person presents you know how their discipline does it that tends to get boring after a while but if you've got a problem to solve for some someone or or or you know a problem then you start to really get to grips with what the different people can bring to the table you know and I've been very impressed for example working with Lenny Smith and Erica Thompson who these weather forecasting mathematician modelers um the way they manag to frame and formulate questions is just fantastic I mean I think models can help you think very clearly about things including thinking where the weaknesses in a particular system are and I think that that's what you could add to the narrative theory is really doing research to try and unpick the detail of the narratives people are putting forward and to understand you know that diagram I put up is a very complex diagram but actually digging into in any given case such as you know the decision to abandon in the UK um particular Railway project or or the decision whether or not to um you know what what we should be doing about Heating and insulation over the you know these kind of big questions you they're bound to be lots of narratives then could you actually look at them very carefully and subject them to to the kind of rigorous thinking that you get in this cognitive modeling and so on that that I think is the

way forward but I think to do that you have to work actually together on the same problem then you learn from each other cool and definitely a theme that on one hand doesn't seem to be the formal kernel of active inference yet we see it arising again and again like for pragmatism to Blossom there has to be a real problem in a real situation and a real approach and it has to be iterated it's not a spectator sport to work um son of spectator sport to be on the field yeah but I think that's a very cool topic um well where do you go from here like what Avenues of research with your um with Carl Friston's via negativ uh endorsement where do you see this line of work heading I I think it's got quite a lot of I mean there obviously there's room in economics there's also room in what was mentioned about um Futures and foresight science um and what I hope is people will use the framework to do empirical work I mean I I don't think you can really ever imagine asking the question is conviction narrative theory true or false I mean it it seems to me with most high level theories of that sort what it it it usefully is an orientation onto a problem or a set of problems which is very different than standard Theory it doesn't mean standard Theory should be replaced by conviction narrative Theory but it means that the range of questions and research that people might be doing could be considerably broadened I mean that's not a wonderful answer but that that's basically uh I I think the other problem is that to do this stuff properly you need you know I've had in my team two or three psychologists a computer scientist and mathematician and a um and and a social anthropologist but I've never had enough resources to have them all at once and that's the problem of interdisciplinary research you actually need a bigger team and and then the team has to work together which is also complicated so um I I would say that's another very important implication that that it's too easy at the moment to work in a um in a Dugout or in a in a what you said there about conviction narrative Theory it can't be simply validated or invalidated by empirical evidence it's a lot like active inference active inference isn't waiting to be established or disestablished based upon a measurement from a real system however you described conviction narrative Theory as an orienting tool we could even think about that in terms of the UDA Loop like observe Orient okay we Ober OB erve the publication we've oriented to it we've updated our cognitive model and now the decision and the action are where the specifics come into play because someone could deploy narrative conviction and then say that this entity has this conviction and that may be empirically true or false and so it's like all of the validation or invalidation is in the realization exactly but the concept is something that kind of sits before the realization and so that when when it is properly understood and respected for being theoretical then it can be like this Fountain of real testable

pragmatic hypotheses real value being layered over other results and models um yet not an account or an explanation in any specific setting by itself but it's something that lets us jump off of that orienting platform into all these settings that we've just discussed ranging from like the pure Financial investing which most people are not going to directly involve in to the the transport policy and and climate issues so it it has a real scope and it's really cool just to even hear a little bit about how it's developed over the years and during a time of uh at UCL and Beyond development of the cognitive Sciences I mean there's one other obvious application which may concern all of us is to research policy research applications and how how research funding is done um because I think it can be applied you know because basically as as we all know you if you're giving out the money if you're the one giving out the money you've got to develop a conviction narrative that giving money to X is going to do y is going to achieve whatever you're trying to achieve right and at the same time you can see that what x is doing is trying to develop a conviction narrative to persuade you right now one of the things about many funding councils at the moment is that they typically talk about doing Innovative work high risk you know the various phrases high risk Innovation Etc and then what they basically do is give the money to people who look reliable with a good track record which are probably contradictions in uh contradictory activity which isn't to say you shouldn't give money to the usual people but it there's clearly a problem I mean I for example on the book I put my slide up minding the markets you'll see that it was very that George aof as a Nobel prizewinning Economist was kind enough to endorse it and when I was talking with him uh he said that his Nobel paper had been turned down by three journals I think it was and when he got to the dinner for the Nobel laureates there were three or four others at the table who had all had the same experience so of course there are lots of bad ideas that don't get funded but the issue of how people develop conviction which is really how do they manage doubt and doubt and Poss the possibility your reputation will you know if you if you give money to some incompetent group who waste it all right that your reputation will suffer and we know if you go to whitehall if you go to to government people are terribly worried about failing and one of the things I hope conviction narrative uh Theory would do is to help people to see that if you want to succeed you've actually got to fail if you're always succeeding you can't be trying hard enough I really like this research funding or resource allocation setting there's like these nested conviction narratives like at the largest scale like we believe that funding basic research or fundamental research or translational research is going to be beneficial along these ways it's like we're convinced of that and we're going to we're we're hodling on that and then at the

program level we have a suspicion that this or that aspect of this phenomena is something that the program manager is going to get behind and that the resources are all aligned and then like in the review itself we might as well directly ask like to the proposer what are you convinced of and then not like just what are the risks or uncertainty or what makes you avoidant but like what are you convicted of that makes you charge into that uncertainty and super excited to carry out the research and we want to hear about the fuzziness and uncertainty because we don't even know simple things until after the fact so we don't expect you to have the whole plan but at the same time it's like we do need to know that your team and context is convinced so that you'll continue the approach as uncertainty occurs rather than upfront so much consideration of the uncertainty that you might never begin and then that sets up a narrative where even small experimental perturbations which might be on the path to transformative research that those have a negative veilance rather than like an almost motivational veilance and and that's just like that's one example that could come into play in funding allocation immediately yeah I think I I think there's a lot of you know that's a very practical area where it can be really thought about do you have any final comments no it's been nice talking to you and uh I do have I I I do find active inference very interesting and um it kind of it's at a feeling level it feels right to me we may even use conviction narrative in the act of inference education setting that's like you could do a 100 phds in thermodynamics and you still wouldn't have some bar of gold waiting for you at the end that's like oh now I understand equation 2.6 so what is it then that keeps us adding those layers of paint in our Learning Journey I think it's an it's an excellent thought and approach and sentiment and also if you provide the slides um there will request in the chat then I can post them alongside the video so thank you thank you than you very much Daniel thank you everyone see you bye